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ScienceDirect
European Journal of Integrative Medicine 3 (2011) 29–37
European Journal of INTEGRATIVE MEDICINE
www.elsevier.com/eujim

Original article

Complementary medicine oncology research in the Middle-East: Shifting from traditional to integrative cancer care

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Received 21 November 2010; received in revised form 12 December 2010; accepted 4 February 2011

Abstract

Introduction: Complementary and alternative medicine (CAM) in the Middle-East is rooted in historical and traditional schools of medicine and is frequently used by patients with cancer. There is limited data on the extent of CAM research in Middle-Eastern countries regarding cancer care and potential benefits and risks of CAM modalities.
Methodology: In this study, we searched the Medline database for studies in CAM and cancer care published during the years 2000–2010 and referenced by authors affiliated with Middle-Eastern academic institutions.
Results: We located 116 articles by authors active in 12 countries. Research themes included CAM use in cancer care, aspects of doctor–patient communication in cancer care, studies of CAM education for health care providers, ethics and regulation, and CAM safety and quality assurance. Seventy-eight articles examined specific CAM modalities that include herbal medicine, Anthroposophic medicine, dietary and nutritional therapies, mind-body medicine, acupuncture, homeopathy, yoga, and Shiatsu. Fifty-nine articles focused on various aspects of herbal medicine such as ethno-botanical surveys and reviews (7 articles), in vitro studies (33), animal studies (8), and clinical studies (11).
Conclusions: The Middle East is a fertile arena for CAM research in cancer care. Collaborative research can significantly enrich the quality and impact of such research, based on sharing cultural and traditional knowledge.

Keywords: Integrative medicine; Complementary alternative medicine; Cancer; Chemotherapy; Middle-East; Islamic medicine

Introduction

Traditional medicine in the Middle-East is rooted in the medicinal legacy of the ancient kingdoms of Egypt and Mesopotamia, and the monotheistic religions of Judaism, early Christianity, and Islam, which related both to pragmatic herbal practices and to spiritual healing. The Middle-Eastern medicinal repertory was gradually developed over centuries, influenced by the three major medical cultures of the late first millennium BC in Greece, India, and China. The central geographic location of the Middle-East helped establish its role in the trade of herbs and spices, as well as in the exchange of medicinal knowledge between the three continents. The Middle Ages had witnessed some inspiring achievements of Islamic medicine marked by activities of Arab and Jewish physicians such as Ibn-Sina (980–1037 AD) and Maimonides (1135–1204 AD). Nevertheless, the last centuries have witnessed a limited influence of Middle-Eastern traditional medicine compared to the surge of other traditional medical systems (i.e. traditional Chinese and Indian medicine). In most Middle-Eastern countries relatively few research studies have been published in the arena of complementary and alternative medicine (CAM) as opposed to the productive research activities in neighboring European and Far-East countries. For example, a Medline search using the keyword "CAM" restricted to the years 1994–2004 revealed 40 abstracts published in Israel compared to 216 in the UK and 759 in the

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1876-3820/$ – see front matter
doi:10.1016/j.eujim.2011.02.007

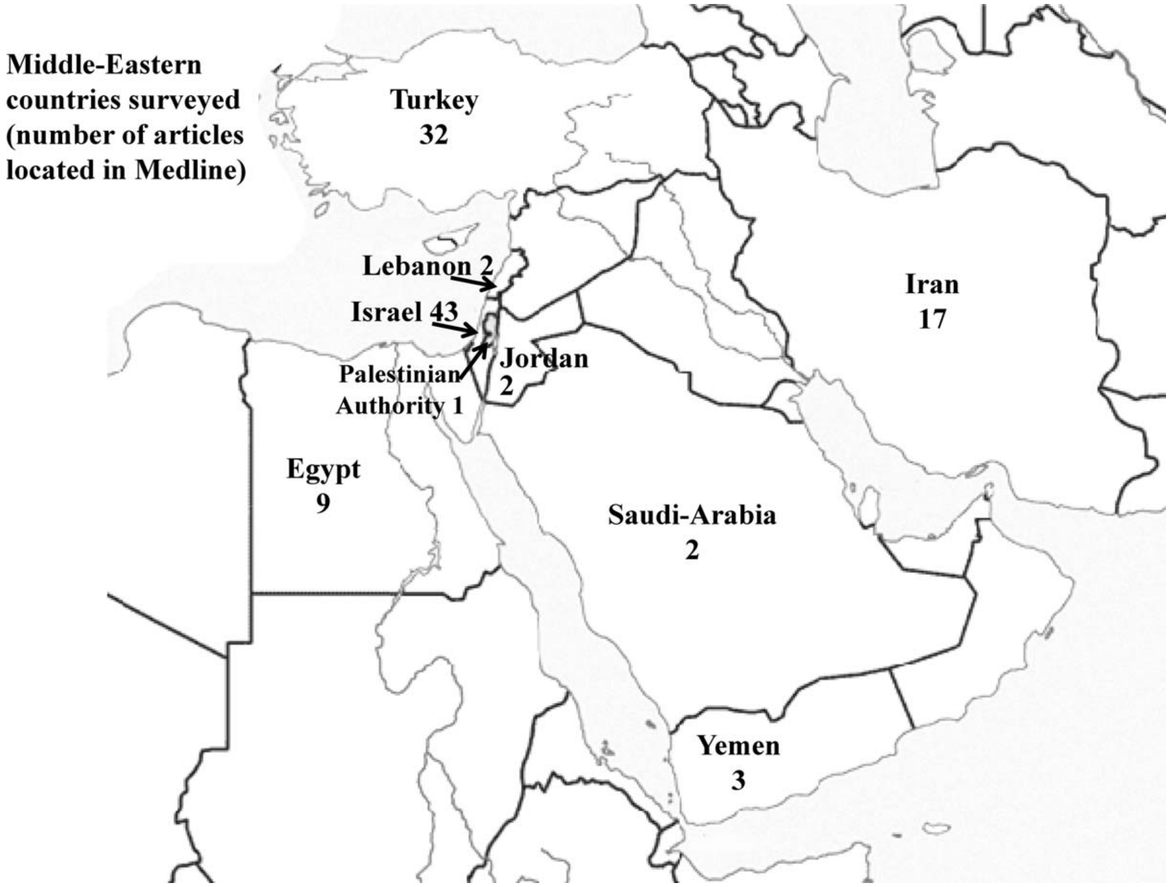

Fig. 1. Middle-Eastern countries surveyed (number of articles located in Medline).

USA [1]. Modern-day prevalence of CAM use in the Middle-East is not fully studied. Ben-Arye et al. compared CAM use in Arab and Jewish communities in primary care clinics in northern Israel and found comparable overall CAM use during the previous year (over 40% of CAM users) [2]. In studies performed in Turkey and Israel, the extent of CAM use in patients with cancer may approach half of the patients [3], even during chemotherapy [4]. This prevalence of CAM use among patients with cancer in the Middle-East is comparable with other countries in the West [5]. Nevertheless, the cross cultural mosaic of societies and religions in the Middle-East and their affiliations with diverse traditional schools of medicine is a potential area for research and contemplation. The use of CAM in cancer care poses several questions: to what degree traditional medicine is interweaved with more “modern” and contemporary CAM modalities? Is CAM use in the Middle-East, especially herbal medicine, limited to indigenous plants and local practices or is it influenced by European, African, and East Asian herbal pharmacopoeia? Can CAM improve cancer care in the Middle-East and what are the constructive routes for its integration in conventional settings? In this paper, we review cancer-related CAM research activities in the Middle-East aiming to document current trends in CAM oncology research in the Middle-East, hoping to inspire multi-disciplinary and multi-national collaboration.

Methods and materials

A multi-disciplinary team of researchers designed the study. The research team included family medicine and internal medicine specialists who practice complementary and integrative medicine, and an Islamic medical historian.

Authors independently searched Pubmed for articles published during the years 2000–2010 using keywords (alone and in various combinations) in the following categories:

a. Oncology related keywords (cancer, oncology, palliative, chemotherapy, and radiation).
b. CAM related keywords (CAM, complementary/alternative/integrative medicine, integrative oncology, Islamic medicine, traditional medicine, ethno-botanical survey, herbs, herbal, mind-body, relaxation, meditation, guided imagery, hypnosis, homeopathy, acupuncture, nutritional/dietary supplements, naturopathy, energy, healing, manual healing, massage, reflexology, yoga, Alexander, Feldenkreis, Anthroposophic medicine, mistletoe, viscum album).
c. Names of countries in the Middle-East (the survey includes states from Morocco in the West to Iran in the East, from Turkey in the North to Yemen in the South).

Table 1
General cancer-related CAM research themes in Middle-Eastern countries.

Research theme	Sub-themes	References (first author, country, year of publication)
CAM use in cancer care (Prevalence of CAM use and patients' expectations)	General oncology	Tarhan, Turkey, 2009 [8]
		Malak, Turkey, 2009 [9]
		Can, Turkey, 2009 [10]
		Ucan, Turkey, 2008 [11]
		Er, Turkey, 2008 [12]
		Montazeri, Iran, 2007 [13]
		Ben-Arye, Israel, 2006 [4]
		Aslan, Turkey, 2006 [14]
		Inanç, Turkey, 2006 [15]
		Tas, Turkey, 2005 [3]
		Pud, Israel, 2005 [16]
		Geva, Israel, 2005 [17]
		Algier, Turkey, 2005 [18]
		Tas, Turkey, 2005 [3]
		Isikhan, Turkey, 2005 [19]
		Gözüm, Turkey, 2003 [20]
		Ceylan, Turkey, 2002 [21]
		Paltiel, Israel, 2001 [22]
	Gynecological cancer	Kav, Turkey, 2008 [23]
		Akyuz, Turkey, 2007 [24]
		Yildirim, Turkey, 2006 [25]
	Pediatric cancer	Genc, Turkey, 2009 [26]
		Gözüm, Turkey, 2007 [27]
		Karadeniz, Turkey, 2007 [28]
		Ben Arush, Israel, 2006 [29]
	Patients with metastatic cancer	Yildirim, Turkey, 2010 [30]
	Palliative care	Yildirim, Turkey, 2006 [31]
	Head and neck	Buentzel, Israel, 2006 [32]
		Talmi, Israel, 2005 [33]
		Aksu, Turkey, 2008 [34]
Doctor–patient communication regarding CAM in oncology setting		Ben-Arye, Israel, 2006 [5]
		Ben-Arye, Israel, 2007 [35]
		Frenkel, Israel, 2005 [36]
		Ben-Arye, Israel, 2004 [37]
CAM education in cancer care	During radiation treatment	Ben-Arye, Israel, 2008 [38]
		Akyuz, Turkey, 2007 [24]
		Ben-Arye, Israel, 2004 [39]
Ethics and regulation in oncology and CAM		Barak, Israel, 2008 [40]
		Ben-Arye, Israel, 2008 [41]
CAM safety and quality assurance		Bogusz, Saudi-Arabia, 2002 [42]
Psycho-social aspects of CAM use in oncology		Ben-Arye, Israel, 2006 [4]
		Montazeri, Iran, 2005 [43]
		Gilbar, Israel, 2001 [44]

Results

Our search identified 116 articles referenced in Medline by authors from Middle-Eastern countries in topics related to CAM and cancer care. Authors were affiliated with academic institutions in the following countries: Israel (43 articles), Turkey (32), Iran (17), Egypt (9), Morocco (3), Yemen (3), Saudi-Arabia (2), Jordan (2), Lebanon (2), Palestinian Authority (1), Tunisia (1), and Kuwait (1) (see Fig. 1). Research themes specified in Table 1 include the following categories: CAM use in cancer care (prevalence of CAM use and patients' expectations) (30 articles), aspects of doctor–patient communication in cancer care (4 articles), studies of CAM education for health care providers (3 articles), ethics and regulation (2 articles), psycho-social aspects (3 articles), and CAM safety and quality assurance (1 article).

Seventy-eight articles examined specific CAM modalities that include herbal medicine, acupuncture, Anthroposophic medicine, dietary and nutritional therapies, homeopathy, mind-body medicine, yoga, and Shiatsu. Table 2 summarizes studies related to herbal medicine in the oncology arena. Fifty-nine articles relate to studies of herbs including ethno-botanical surveys and reviews (7 articles), in vitro studies (33), animal studies (8), and clinical studies (11, including 2 randomized controlled trials). Studies in herbal medicine in the context of cancer are reported concerning 29 single herbs, 4 herbal formulas, 2 herbal-animal mixed formulas, one mushroom-based study, and 3 studies regarding isolated active ingredients of

Table 2
Cancer-related herbal research in the Middle-East.

Sub-themes	Research category	References (first author, country, year of publication)
General herbal medicine studies	Ethno-botanical survey	Said, Israel, 2002 [45]
		Ali-Shtayeh, Palestinian Authority, 2000 [46]
	In vitro studies	Monthana, Yemen, 2009 [47]
		Amirghofran, Iran, 2006 [48]
Allium sativum (Garlic)	Review	Thomson, Kuwait, 2003 [49]
Aloe vera; Ehrlich ascites tumors	Animal study	Akev, Turkey, 2007 [50]
Centaurea ainetensis; Salograviolide A; colon cancer	In vitro study	El-Najjar, Lebanon, 2008 [51]
Citral	In vitro study	Dudai, Israel, 2005 [52]
		Chaouki, Morocco, 2009 [53]
Coriolus versicolor and Funalia trogii	In vitro study	Unyayar, Turkey, 2006 [54]
Crocus sativus L. (Saffron); transitional cell carcinoma	In vitro studies	Tavakkol-Afshari, Iran, 2008 [55]
		Feizzadeh, Iran, 2008 [56]
Curcumin (Curcuma longa, Turmeric); pancreatic cancer; colon cancer; head and neck cancer; doxorubicin-induced toxicity	In vitro studies	Lev-Ari, Israel, 2007 [57]
		Lev-Ari, Israel, 2006 [58]
		Lev-Ari, Israel, 2006 [59]
	Animal study	Mohamad, Egypt, 2009 [60]
	Review	Bar-Sela, Israel, 2010 [61]
		Khafif, Israel, 2010 [62]
Cyperus rotundus	In vitro study	Kilani-Jaziri, Tunisia, 2009 [63]
Daphne gnidium L.; breast cancer	In vitro study	Chaouki, Morocco, 2009 [64]
Dionysia termeana	In vitro study	Amirghofran, Iran, 2007 [65]
Ficus carica (Fig); Cancer cell proliferation	In vitro study	Rubnov, Israel, 2001 [66]
Glycyrrhiza glabra (Licorice); Glabridin	In vitro study	Tamir, Israel, 2000 [67]
Haplophyllum canaliculatum Boiss	In vitro study	Varamini, Iran, 2009 [68]
Honey; mucositis; head and neck cancer	Clinical study – RCT	Rashad, Egypt, 2008 [69]
		Motallebnejad, Iran, 2008 [70]
Hypericum adenotrichum	In vitro study	Ozmen, Turkey, 2009 [71]
Lawsonia inermis (Henna)	Clinical study – non RCT	Yucel, Turkey, 2008 [72]
Linum persicum and Euphorbia cheiradenia; leukemia cell lines	In vitro study	Amirghofran, Iran, 2006 [73]
Morinda citrifolia L. (Noni)	Animal study	Taşkin, Turkey, 2009 [74]
Nerium oleander; Leukemia	In vitro study	Turan, Turkey, 2006 [75]
Nigella sativa (Black cumin)	Animal studies	El-Aziz, Egypt, 2005 [76]
		Salim, Egypt, 2003 [77]
		Mabrouk, Egypt, 2002 [78]
	In vitro study	Abuharfeil, Jordan, 2007 [79]
		El-Obeid, Saudi-Arabia, 2006 [80]
	Review	Edris, Egypt, 2009 [81]
Peganum harmala	In vitro study	Lamchouri, Morocco, 2000 [82]
Punica granatum (pomegranate)	In vitro study	Lansky, Israel, 2005 [83]
	Review	Lansky, Israel,2007 [84]
Ruta graveolens	In vitro study	Varamini, Iran, 2009 [85]
Salix safsaf (willow); human leukemic cells	In vitro study	El-shemy, Egypt, 2003 [86]
Salvia disermas (Sage)	In vitro study	Hawas, Egypt, 2009 [87]
Salvia libanotica (Sage); colon cancer	In vitro study	Itani, Lebanon, 2008 [88]
Scutellaria orientalis; Leukemia	In vitro study	Ozmen, Turkey, 2010 [89]
Teucrium polium L.	In vitro study	Rajabalian, Iran, 2008 [90]
Triticum aestivum (Wheat grass juice); hematological toxicity; breast cancer	Clinical study – non RCT	Bar-Sela, Israel, 2007 [91]
Vaccinium oxycoccos (Cranberry juice); lymphoma; mice	Animal study	Hochman, Israel, 2008 [92]
Viscum Album; ascites; advanced cancer; colorectal cancer; hepatocellular carcinoma	Clinical studies – non RCT	Bar-Sela, Israel, 2006 [93]
		Mabed, Egypt, 2004 [94]
		Bar-Sela, Israel, 2004 [95]
Middle-Eastern herbal formulas		
Iranian herbs (Galium mite, Ferulago angulata, Stachys obtusicrena, Cirsium bracteosum and Echinophora cinerea)	In vitro study	Amirghofran, Iran, 2006 [48]
HESA-A herbal-marine formula [Penaeus latisculatus (king prawn), Carum carvi, and Apium Graveolens]; end-stage metastatic cancer; colon cancer; breast cancer	Clinical studies-non RCT's	Ahmadi, Iran, 2010 [96]
		Ahmadi, Iran, 2009 [97]
		Ahmadi, Iran, 2005 [98]
		Ahmadi, Iran, 2005 [98]
Yemen herbs (Ballochia atro-irgata, Eureiandra balfourii and Hypoestes pubescens)	In vitro study	Mothana, Yemen, 2009 [99]
		Monthana, Yemen, 2009 [100]
Chinese herbal medicine		
Chinese herb formula, anti-cancer number one	Animal study	Hong-Fen, Israel, 2001 [101]

Table 3
Cancer-related research in the Middle-East in CAM modalities other than herbal medicine.

Research theme	Sub-themes	Research category	References (first author, country, year of publication)
General CAM overview		Review	Ben-Arye, Israel, 2004 [102]
Acupuncture	Pain, nausea, breathlessness, vasomotor symptoms and limb edema	Review	Samuels, Israel, 2002 [103]
Anthroposophic medicine	Viscum Album; ascites; advanced cancer; colorectal cancer; hepatocellular carcinoma	Clinical studies – non RCT	Bar-Sela, Israel, 2006 [92]
			Mabed, Egypt, 2004 [93]
			Bar-Sela, Israel, 2004 [94]
	Art therapy; fatigue; depression	Clinical study – non RCT	Bar-Sela, Israel, 2007 [104]
Diet and nutritional supplements	Nutritional supplements counseling	Review	Alsawaf, Jordan, 2007 [105]
			Frenkel, Israel, 2005 [34]
			Frenkel, Israel, 2002 [106]
	Kefir supplementation; colorectal cancer	Clinical study – RCT	Can, Turkey, 2009 [107]
	Tomato lycopene; colon cancer	Clinical study – RCT	Walfisch, Israel, 2007 [108]
Homeopathy	Stomatitis; hemato-oncology	Clinical study – RCT	Oberbaum, Israel, 2001 [109]
Mind body medicine	Progressive Muscle Relaxation with Guided Imagery	Clinical studies – RCT non RCT	Baider, Israel, 2001 [110]
			Baider, Israel, 1994 [111]
	Hypnosis; xerostomia; head and neck cancer	Clinical study – non RCT	Schiff, Israel, 2009 [112]
	Guided imagery; Xerostomia, Anthroposophic, Ayurveidic and Chinese medicine	Theoretical study	Ben-Arye, Israel, 2009 [113]
	Mind-body; ovarian carcinoma	Case report	Lev-Ari, Israel, 2006 [114]
	Prayer during chemotherapy	Cross sectional survey	Rezaei, Iran, 2008 [115]
Shiatsu	Supportive care	Review	Argash, Israel, 2009 [116]
Yoga	Breast cancer; quality of life	Clinical study – non RCT	Ulger, Turkey, 2010 [117]

herbs. Most-researched herbs, in terms of the number of articles, were Curcuma longa (Turmeric, 6 articles), Nigella sativa (Black cumin, 6 articles), Viscum Album (Mistletoes, 3 articles), and the HESA-A herbal-marine formula (4 articles).

Table 3 summarizes studies in CAM modalities other than herbal medicine. Specific CAM modalities included mind-body medicine (6 articles), dietary and nutritional therapies (5), Anthroposophic medicine (4 articles), acupuncture (1), homeopathy (1), yoga (1), and Shiatsu (1).

Discussion

Our review shows vivid research activity of CAM research in the oncology field by researchers all over the Middle-East. This rich diversity of research productivity is intriguing since no national CAM research institution exists in any of the countries surveyed. Nevertheless, the research potential of the region in integrative oncology is promising and indicates the need to establish a collaborative Middle-East integrative oncology research center. In general, the points of strength in contemporary Middle-Eastern cancer-related CAM research are in the field of herbal medicine and in surveys aimed to study patients' use of CAM in the context of cancer, their perspectives and motives and the influence of CAM use in terms of doctor–patient communication and cancer care. Our review shows that herbal medicine is the most reported modality in publications regarding CAM use among patients with cancer in the Middle-East, followed by nutritional and mind-body and spiritual practices. The proportion of published data on herb in cancer care may manifest its increased use by people with cancer, similar to patterns of CAM use in Europe [6]. The high usage of herbs in the Middle-east may originate from the tradition of Greco-Islamic medicine, which needs to be regarded as important historical and ethno-botanical source for further studies focusing in cancer care. The importance of this herbal tradition is also emphasized in our review by the high number of publications in herbal medicine compared to other CAM modalities. We suggest that there is a need for further research aimed at tracing specific plants which Middle-Eastern patients actually use during cancer treatment. Subsequently, in vitro and clinical studies should test the efficacy and safety of these herbs including possible interactions with chemotherapy and other conventional drugs. Herbal research might also explore clinical benefits of herbs referenced for cancer care in historical Islamic and Jewish medical literature and ethno-botanical surveys. We suggest that future traditional medicine literature surveys should inform us regarding non-herbal CAM modalities such as mind-body medicine (e.g. breathing exercises, hypnosis-like treatments), spiritual practices (e.g. magic, enchanting, Dervish dance), touch practices (e.g. reflexology, massage), and manual modalities that resemble Far-Eastern and Ayurvedic medicine (e.g. use of cupping, medicinal oils, acupressure-like "energy" points).

Our review is limited by a search strategy that does not necessarily assess the full volume of Middle-East CAM research in cancer-related themes. We have encountered several manuscripts that were not coded with key words related to CAM or herbal medicine. In other cases, we traced manuscripts on the basis of professional or personal acquaintance with specific researchers and their colleagues. This should not be considered only as "selection bias" but as a complementary root to locate researchers that do not necessarily present their studies in scientific meetings in the West. Language bias is another limitation that should be considered in terms of non-English publications (e.g. in Arabic, Turkish, and Hebrew) as well as the potential difficulty some researchers have in writing English which may limit their ability to publish articles in high impact factor journals. In order to limit this publication's bias, we suggest setting up a multi-national registration system for documenting all past and present research conducted in the Middle East in the integrated field of CAM and oncology. Based on the current promising data, we call for a collaborative forum of multi-disciplinary researchers from Middle-Eastern countries that would locate articles and research the same subjects in each country. This forum could encourage collaborative studies on common research themes. For example, the Nigella sativa plant, which was independently studied in three countries (Egypt, Jordan, and Saudi-Arabia – see references in Table 2), is a potential candidate for a multi-national study in several countries throughout the Middle-East. Another task of the Middle-East data-registration system for integrative oncology research would to be locate and identify research need in specific countries based on findings in other countries. For example, most of the studies on CAM use in cancer care were performed in Turkey and Israel (see Table 1) but knowledge is lacking as to the prevalence of CAM use and patients' expectations in other countries. A Middle-East data-registration system in integrative oncology may be supported by collaboration with other registration systems, especially in Europe, such as the Concerted Action for Complementary and Alternative Medicine Assessment in the Cancer Field (CAM-Cancer) hosted by the National Information Center for Complementary and Alternative Medicine in Norway [7].

In conclusion, based on our findings we suggest further in-depth research based on the collaborative teamwork of researchers across the Middle-East with the goal of documenting CAM studies in a Middle-Eastern integrative oncology research database. More research is warranted to understand the clinical impact of integration in real life clinical practice.

Conclusions

The Middle East is a fertile arena for CAM research in cancer care. Collaborative research can significantly enrich the quality and impact of such research, based on sharing cultural and traditional knowledge.

Conflict of interest

No conflict of interest declared.

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